

Walls or cascade? A pre-registered test of wall dissipation in decaying magnetohydrodynamic turbulence

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(Dated: 2 October 2026)

Does the small-scale dissipation of magnetohydrodynamic (MHD) turbulence come from the cascade in the bulk, as in a periodic box, or from the walls that confine laboratory plasmas and liquid metals? We test this with a controlled numerical experiment. The symmetries of the Taylor–Green MHD flow make the box $[0, \pi]^3$ a closed domain with free-slip, perfectly conducting walls: solving that box directly reproduces the periodic flow to round-off, and making its walls no-slip, with everything else unchanged, isolates what the walls do. With two independent lattice Boltzmann implementations we follow the decay at unit magnetic Prandtl number for Reynolds numbers 250 to 2000, on grids that resolve the $\text{Re}^{-1/2}$ wall layer, with the observable, the resolution criterion and the decision rule fixed before the runs. No-slip walls add dissipation within $\text{Re}^{-1/2}$ of them at every Reynolds number and under every definition we tried. The excess is viscous: the near-wall Ohmic dissipation is 40–45% lower than with free-slip walls. Snapshots of the local dissipation show current sheets crossing the near-wall layer of the free-slip box and a broad viscous layer at the no-slip walls. The pre-registered reading, the excess at the dissipation peak, falls from 0.063 to 0.036 between $\text{Re} = 1000$ and 2000, the outcome we had called “cascade”. That fall is set by which of two maxima of the dissipation is the larger: the peak switches between rungs, and over a factor of eight in Reynolds number the trend of the wall share depends on when it is measured, while its sign does not.

I. INTRODUCTION

Decaying magnetohydrodynamic (MHD) turbulence has been studied mostly in periodic boxes.^{1–5} There the dissipation is made in current and vorticity sheets that the cascade forms in the bulk, and its rate tends to a value independent of the Reynolds number, Re ,^{1,4,5} the MHD counterpart of the finite dissipation of hydrodynamic turbulence.⁶

The flows that motivate much of this work, in liquid metals and laboratory plasmas, are bounded, and walls add boundary layers that dissipate. Whether that dissipation survives as the viscosity vanishes is a question about the walls themselves. Kato showed that a Navier–Stokes flow in a bounded domain converges to its Euler limit if and only if the dissipation in a layer of thickness proportional to the viscosity vanishes.⁷ In two-dimensional hydrodynamics, Nguyen van yen, Farge and Schneider found dissipating structures produced at no-slip walls whose dissipation appears to tend to a finite value as Re grows.⁸ Confined MHD turbulence has been simulated with volume penalisation in two⁹ and three¹⁰ dimensions. We are not aware of a comparison that changes only the wall condition and follows the answer as Re increases.

We therefore ask: in decaying three-dimensional MHD turbulence, does a no-slip wall take a share of the dissipation that holds as Re grows (“walls”), or does that share fall, so that the periodic picture of a bulk cascade survives confinement (“cascade”)?

The answer rests on a construction that isolates the walls. The Taylor–Green (TG) MHD flows of Lee *et al.*² and Pouquet *et al.*,³ which generalize the hydrodynamic TG vortex,¹¹ are symmetric about the faces of $[0, \pi]^3$. On those faces the velocity is impermeable and free-slipping, and the magnetic field has the parity of a perfect conductor or, for a second field, of a pseudo-vacuum. The periodic flow restricted to $[0, \pi]^3$ is therefore exactly the flow in a closed box with those walls. We solve that box directly, with walls on lattice nodes, and check that it reproduces the periodic flow to round-off. We then change the velocity condition from free slip to no slip and nothing else. The difference between the two boxes is the effect of the walls.

A question with two outcomes invites an analysis flexible enough to find either. We therefore fixed the observable, the resolution criterion and the decision rule in version-controlled job scripts before the runs they govern,^{12,13} and we report them as fixed. Section II describes the problem, the method and the protocol, Sec. III the verification, Sec. IV the results, and Sec. V what they settle and what they do not. In brief, the no-slip excess is robust: it is positive at every Re and under every definition we tried, it is viscous, and it comes with less near-wall Ohmic dissipation than free slip. The pre-registered verdict is “cascade”. But that verdict rests on a switch between two maxima of the dissipation, and over the factor of eight in Re that one GPU affords, the trend of the wall share depends on when it is measured.

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II. PROBLEM, METHOD AND PROTOCOL

A. The flow and the box

We solve the incompressible visco-resistive MHD equations in units where the initial velocity amplitude and the density are one,

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + (\nabla \times \mathbf{B}) \times \mathbf{B} + \nu \nabla^2 \mathbf{v}, \quad (1)$$

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \quad (2)$$

with $\nabla \cdot \mathbf{v} = \nabla \cdot \mathbf{B} = 0$ and $\nu = \eta = 1/\text{Re}$, so the magnetic Prandtl number is one and $\text{Re} = 1000$ is run C2 of Ref. 3. The initial conditions are those of Refs. 2 and 3:

$$\mathbf{v} = (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0), \quad (3)$$

$$\mathbf{B}_C = b_0 (\sin 2x \cos 2y \cos 2z, \cos 2x \sin 2y \cos 2z, -2 \cos 2x \cos 2y \sin 2z), \quad (4)$$

$$\mathbf{B}_I = b_0 (\cos x \sin y \sin z, \sin x \cos y \sin z, -2 \sin x \sin y \cos z), \quad (5)$$

with $b_0 = 1/\sqrt{3}$, so that the kinetic and magnetic energies, $E_V = \langle |\mathbf{v}|^2 \rangle / 2$ and $E_M = \langle |\mathbf{B}|^2 \rangle / 2$, are both $1/8$ at $t = 0$.

On the six faces of $[0, \pi]^3$ the velocity (3) has $\mathbf{v} \cdot \mathbf{n} = 0$ and $\partial_n \mathbf{v}_t = 0$: an impermeable, free-slip wall. The field $\mathbf{B}_C$ has $\mathbf{B} \cdot \mathbf{n} = 0$ and $\partial_n \mathbf{B}_t = 0$. This is what a perfectly conducting wall imposes, since $\mathbf{n} \times \mathbf{E} = 0$ with Ohm's law $\mathbf{E} = \eta \mathbf{j} - \mathbf{v} \times \mathbf{B}$ and $\mathbf{v} \cdot \mathbf{n} = \mathbf{B} \cdot \mathbf{n} = 0$ gives $\mathbf{n} \times \mathbf{j} = 0$, and it holds with either velocity condition. The field $\mathbf{B}_I$ has $\mathbf{B} \times \mathbf{n} = 0$ and $\partial_n B_n = 0$, the pseudo-vacuum condition. These parities are preserved by Eqs. (1)–(2). The boxes are:

A: free slip, conducting walls, $\mathbf{B}_C$: an exact restriction of the periodic flow, and the control;

B: no slip, conducting walls, $\mathbf{B}_C$: the experiment, which differs from A in the velocity condition only;

C: no slip, no field: the hydrodynamic reference;

A', B': as A and B, with $\mathbf{B}_I$ and pseudo-vacuum walls.

The TG velocity does not vanish on the faces, so a no-slip box starts impulsively: its wall nodes are set at rest at $t = 0$, and a vortex sheet on every wall dominates the early dissipation. In B the dissipation rate at $t = 0$ is $\varepsilon = 0.30, 0.22, 0.15$ and 0.09 at $\text{Re} = 250, 500, 1000$ and 2000 , 95–98% of it within $\text{Re}^{-1/2}$ of the walls.

B. Lattice Boltzmann discretization

The velocity is carried by a D3Q27 distribution with a central-moment collision in a shifted basis,^{14,15} and the Maxwell stress enters its equilibrium.¹⁶ The field is carried by Dellar's vector-valued distribution¹⁶ on D3Q7

TABLE I. Runs: two grids per rung. δ/h is the number of lattice spacings in the layer thickness $\delta = \text{Re}^{-1/2}$; $\tau - \frac{1}{2}$ is the fluid relaxation parameter; steps to $t = 10$ and the largest lattice Mach number over A and B are for the finer grid; “bits” is the floating-point precision.

Re	N	δ/h	$\tau - \frac{1}{2}$	steps	Ma _{max}	bits
125	128, 192	3.62, 5.44	0.0485, 0.0730	12159	0.094	64
250	192, 256	3.85, 5.13	0.0365, 0.0487	16234	0.102	64
500	256, 384	3.63, 5.45	0.0244, 0.0366	24383	0.109	64
1000	384, 512	3.86, 5.14	0.0183, 0.0244	32531	0.117	64
2000	512, 640	3.64, 4.55	0.0122, 0.0153	40680	0.133	32

with a single-relaxation-time collision; ν and η set the two relaxation times.¹⁷ The box has N nodes a side with the walls on nodes 0 and $N - 1$, so the spacing is $h = \pi/(N - 1)$. The lattice velocity of the unit velocity is $u_0 = 0.05$, one step is $\Delta t = u_0 h$, and $\nu_{\text{lat}} = u_0/(h\text{Re})$; the fluid relaxation parameter is $\tau - \frac{1}{2} = 3\nu_{\text{lat}}$, and the magnetic one is $4\eta_{\text{lat}}$ on D3Q7. Table I lists the runs.

All walls are on nodes, and each wall node collides like any other once its unknown populations are set. For free slip these are the mirror images of their partners about the wall (specular reflection). For no slip we use the regularized velocity condition of Latt *et al.*¹⁸ with zero wall velocity; at the twelve edges and eight corners, where it has no density stencil, the density is set to one. For the field we use a parity wall. Write the field component a as carried by its own populations $g_i^{(a)}$. At a node on a wall normal to axis k , every population i entering the box is replaced by

$$g_i^{(a)} \leftarrow s_{ak} g_{i^*}^{(a)}, \quad (6)$$

where i^* is the mirror of i about the wall and $s_{ak} = -1$ if the wall flips component a (the normal one, $a = k$, at a conducting wall; the tangential ones, $a \neq k$, at a pseudo-vacuum wall) and $+1$ otherwise. At edges and corners the mirrors compose and the signs multiply. A conducting wall thus flips only the normal component, and a pseudo-vacuum wall only the tangential ones. Dellar has given moment-based magnetic boundary conditions for the same distribution;¹⁹ Eq. (6) is a population-level mirror that imposes a parity on any combination of faces. Three tests in the repository check it. First, with free slip and either parity, the box of N nodes reproduces the periodic 2π box of $2(N - 1)$ nodes to 10^{-14} , node for node and transient included. Second, $\mathbf{B} \cdot \mathbf{n}$ on the wall nodes stays at 10^{-32} . Third, in the no-slip box the exact decay of the conducting-wall eigenmode $\mathbf{B} = B_0(\sin x \cos y, -\cos x \sin y, 0)$, which leaves the fluid at rest, converges at second order (order 2.00 over $N = 17, 33, 65$).

The method exists in two implementations that share no source code: one in C++20 on Kokkos,^{20,21} one in CUDA. Their host builds agree to every printed digit over 815 steps at $N = 65$, $\text{Re} = 200$, with free and no slip.

The CUDA build on the device agrees with the Kokkos reference to within 6×10^{-16} of each diagnostic's scale. The production runs used NVIDIA A100 (80 GB) GPUs: double precision up to $\text{Re} = 1000$, and single precision at $\text{Re} = 2000$, where $N = 640$ fits in memory only in single precision. The runs reported here took 34 GPU-hours.

C. Diagnostics

The dissipation rate is $\varepsilon = \nu \langle |\boldsymbol{\omega}|^2 \rangle + \eta \langle |\mathbf{j}|^2 \rangle$, with $\boldsymbol{\omega} = \nabla \times \mathbf{v}$ and $\mathbf{j} = \nabla \times \mathbf{B}$. The box average $\langle \cdot \rangle$ uses trapezoidal weights, $\frac{1}{2}$ on a face, $\frac{1}{4}$ on an edge and $\frac{1}{8}$ at a corner, which makes the box average of a mirror-symmetric field equal to its periodic average exactly. Derivatives are second-order central differences, extended at a wall by the wall's own rule: the mirror for a free-slip velocity and for the magnetic parity, and a one-sided second-order stencil for a no-slip velocity.

Each node is at distance $d = kh$ from its nearest wall, $k = 0, 1, \dots$. With the trapezoidal weights, the nodes with $d = kh$ fill exactly the part of the box at distances between $(k - \frac{1}{2})h$ and $(k + \frac{1}{2})h$ from the walls (between 0 and $h/2$ for the wall nodes). If e_k is the dissipation in layer k , the dissipation within a distance rh of the walls is

$$\mathcal{E}(r) = \sum_{j < k} e_j + (r - k + \frac{1}{2}) e_k, \quad k = \lfloor r + \frac{1}{2} \rfloor, \quad (7)$$

that is, linear within each layer (with $\mathcal{E} = 2re_0$ for $r < \frac{1}{2}$). The share of the dissipation within $m\delta$ of the walls, $\delta = \text{Re}^{-1/2}$, is

$$f_m(t) = \mathcal{E}(m\delta/h) / \sum_k e_k, \quad m = 1, 2, 4, \quad (8)$$

and it divides into a viscous and an Ohmic part, $f_m = f_m^\nu + f_m^\eta$, each a share of the total ε . The dissipation density relative to the box mean is a layer's share of ε over its share of the volume.

Counting whole node layers instead (every node with $kh < m\delta$) measures a layer of thickness $(\lceil m\delta/h \rceil - \frac{1}{2})h$ rather than $m\delta$: 0.91 δ at $N = 384$ and 1.07 δ at $N = 512$ for $\text{Re} = 1000$. In our first set of ladder runs this made f_1 differ by 8–25% between the two grids of a rung while ε agreed to 1%. Swept across $N = 33$ to 49 at fixed physics, the count makes f_4 jump between 0.52 and 0.60 each time $4\delta/h$ crosses a node, while Eq. (7) keeps it within 0.561–0.566. Any layer, shell or threshold observable on a lattice can do this.

We also use, after the fact (Sec. IV F), the time-integrated share over a window $[t_0, t_1]$,

$$F_m = \int_{t_0}^{t_1} f_m \varepsilon dt / \int_{t_0}^{t_1} \varepsilon dt, \quad (9)$$

the fraction of all the energy dissipated in the window that was dissipated within $m\delta$ of the walls. It is evaluated

with the trapezoidal rule over the probes, which are 0.05 or 0.1 apart, with every integrand interpolated to the window's edges.

D. The pre-registered protocol

The two outcomes of Sec. I were stated, qualitatively, on 25 September 2026, before any production run. The following rules were committed with the job scripts before the runs they govern. The observable and the gate were committed with the ladder on 27 September 2026. The estimator of f_m was corrected on 28 September, after the first ladder failed the gate for the reason given above; the observable it estimates and the gate did not change. The Mach control was fixed on 29 September, and the rule for the top rung on 30 September. Both came after the lower rungs had been run and before the runs they judge.

1. *Observable*: B–A in f_1 at each run's turbulent dissipation peak. The peak is the maximum of ε after ε has first risen 5% above its running minimum, refined by a parabola through the three probes around the discrete maximum; f_m is interpolated to that time. The rule skips the impulsive start of a no-slip box; for free slip the peak is the global maximum. A run in which ε never rises 5% above its running minimum has no turbulent peak and no f_m .
2. *Resolution gate*: at every (box, Re), f_1 , f_2 and f_4 at the peak agree within 5% (relative) between the two grids.
3. *Band*: the grid differences of f_1 for A and for B, added.
4. *Decision*: “walls hold” if B–A at the top rung lies within the combined bands of its value at the rung below; otherwise “cascade”.
5. *Controls*, each changing one thing: u_0 halved at $\text{Re} = 500$ and 1000, which halves the Mach number and $\tau - \frac{1}{2}$ together; and double against single precision at $\text{Re} = 2000$, $N = 512$. A control passes if it moves B–A by less than the band.

The time-integrated share (9) was not part of the protocol; we introduce it in Sec. IV F because of what Sec. IV E found.

III. VERIFICATION

The free-slip box A at $\text{Re} = 1000$ is run C2 of Pouquet *et al.*³ Its minimum ratio of magnetic to kinetic energy is $E_M/E_V = 0.3695$ and 0.3693 at $N = 384$ and 512, reached at $t = 7.3$, against their 0.35. The first maximum of Ω_M/Ω_V after its initial fall is 2.154 and 2.159, against their “2.”. Here Ω_V and Ω_M are the kinetic and

magnetic enstrophies. Our ratio is converged to four figures between the two grids and lies 5.5% above theirs, which is quoted to two figures. A run at $N = 97$ gave 0.357, closer to their value but at $t = 9.8$ on an under-resolved grid.

Table II shows the resolution gate. Every rung of A and B with a turbulent peak passes it, the worst difference being 3.0% in f_1 of A at $\text{Re} = 2000$. Without a turbulent peak by $t = 10$ on either grid are B at $\text{Re} = 125$ and C at $\text{Re} \leq 500$; C at $\text{Re} = 2000$ is still rising at $t = 10$. Halving u_0 moves B–A by -0.0014 at $\text{Re} = 500$ and by -2×10^{-5} at $\text{Re} = 1000$, against bands of 0.0030 and 0.0004. Double precision at $\text{Re} = 2000$ moves it by less than 10^{-5} against 0.0021, although the two runs differ by up to 4×10^{-5} (relative) in their time series. All three controls pass. Over every run, the total mass drifts by at most 4.4×10^{-5} (relative), and the r.m.s. of $\nabla \cdot \mathbf{B}$ stays below 1% of the r.m.s. of $|\mathbf{j}|$. The largest lattice Mach number is 0.133, in A at $\text{Re} = 2000$ (Table I).

IV. RESULTS

A. Where the dissipation sits

Figure 1(a) shows how the dissipation is spread across the box at $\text{Re} = 1000$, integrated over $t = 2$ –10. At the walls, the dissipation density is 8.50 times the box mean in C, 2.43 times in B and 0.38 times in A. The free-slip conducting walls are quieter than the bulk. The no-slip walls of the MHD box are more active than the bulk, but much less so than those of the hydrodynamic box. In C the density falls below the mean beyond $d \approx 4\delta$: in that box the walls do most of the dissipating. Over $t = 2$ –10, C dissipates 33% of its energy within δ of the walls and B 9.9%. At its own turbulent peak ($t = 3.20$), C's f_1 is 0.50. C carries half of B's initial energy, since it has no field. This is the one rung at which both boxes have a turbulent peak, so it is the one rung at which they can be compared.

Figure 1(b) follows the density at the walls with Re. In B it rises monotonically, 1.44, 1.82, 2.43 and 3.13 times the mean at $\text{Re} = 250$ –2000. In A it stays near 0.38 and falls to 0.34 at $\text{Re} = 2000$. B's wall density rises monotonically in each of the five time windows of Sec. IV F, as $\text{Re}^{0.25}$ to $\text{Re}^{0.44}$ depending on the window. A's changes as $\text{Re}^{-0.02}$ to $\text{Re}^{-0.11}$.

B. The no-slip excess and what it is made of

Table III and Fig. 2(b) give the time-integrated share F_1 over $t = 2$ –10. No-slip walls take more of the dissipation next to them at every rung: B–A = 0.059, 0.055, 0.054 and 0.048 at $\text{Re} = 250$ –2000, with grid bands of at most 0.0009. The excess is viscous. Its viscous part is +0.072 to +0.059 and its Ohmic part is *negative*, -0.014 to -0.011 . That deficit is not an artefact of dividing by a

larger total. Over the window the no-slip box dissipates *less* energy than the free-slip one (0.55–0.72 of the initial energy, against 0.62–0.75), so its near-wall Ohmic dissipation is lower in absolute terms too, 0.55–0.59 of A's. In A the Ohmic part is the larger of the two near the walls: 0.039 against 0.030 at $\text{Re} = 250$, and 0.026 against 0.012 at $\text{Re} = 2000$.

C. What the fields show

The shares above are integrals of local fields. Figures 3 and 4 show the fields themselves at $\text{Re} = 1000$: at $t = 2.3$, B's turbulent peak, and at $t = 4.6$, by the later maxima. They come from reruns of the production runs that also write the full fields, and the reruns reproduce the production time series exactly, at every one of the 48 probes to $t = 4.7$. The local viscous and Ohmic dissipation is computed from the fields with the drivers' own derivatives at the walls; summed over the box, it reproduces ε to 2×10^{-8} (checked at $N = 65$ for all three boxes).

In the bulk the two MHD boxes are alike. In the section $y = \pi/4$ they show the same current and vorticity sheets, and their plane means differ by less than 15%: viscous 0.37 and 0.39 of the box mean in A and B at $t = 2.3$, Ohmic 0.49 and 0.42. They differ next to the walls. In the plane 0.58δ above the bottom wall, B's viscous dissipation is broad and strong, 1.49 of its box mean against A's 0.53, and its Ohmic dissipation is weaker, 0.39 against 0.61. In absolute terms, B's viscous dissipation there is 3.5 times A's and its Ohmic 0.79 of A's. In A the plane is crossed by current sheets, and bright ones run along its edges, where it meets the vertical walls. By $t = 4.6$ the bulk of both boxes is more finely structured and A's near-wall plane has quietened, but the contrast remains: B's viscous dissipation there is 6.8 times A's and its Ohmic 0.68 of A's. These are single planes at two instants, but they point the same way as the time-integrated profiles of Sec. IV B, in which B's near-wall Ohmic dissipation is 0.55–0.59 of A's.

The box without a field is different in kind. Its sections glow at all four walls while its bulk is nearly uniform, and its near-wall plane dissipates at 6.7 times its box mean at $t = 2.3$ and 3.3 times at $t = 4.6$, about 3.5 and 3.2 times B's relative level.

The near-wall planes show the bottom wall. By the symmetries of the TG flow the six walls fall into two classes, the two horizontal walls and the four vertical ones, and these planes show only the first; the sections cross both.

D. The pre-registered verdict

Read at the turbulent peak, as registered, B–A is 0.0732, 0.0648, 0.0626 and 0.0361 at $\text{Re} = 250$ –2000 [Fig. 1(c)], with bands of 0.0020, 0.0030, 0.0004 and 0.0021. From $\text{Re} = 1000$ to 2000 it falls by 0.0266, 10.5

TABLE II. Resolution gate and controls. Relative differences of f_1 , f_2 and f_4 at the turbulent peak between the two grids of each rung, the band of B–A (the two grid differences of f_1 , added), and the shift of B–A in the control run: Mach, u_0 halved; FP64, double precision at $N = 512$.

Re	N	A: $\Delta f_1/\Delta f_2/\Delta f_4$ (%)	B: $\Delta f_1/\Delta f_2/\Delta f_4$ (%)	band	control shift
250	192, 256	0.7/0.2/0.1	1.0/0.7/0.2	0.0020	–
500	256, 384	0.8/0.5/1.0	2.1/1.5/0.7	0.0030	Mach, -0.0014
1000	384, 512	0.7/0.7/0.3	0.2/0.1/0.2	0.0004	Mach, $< 10^{-4}$
2000	512, 640	3.0/1.3/0.8	0.5/0.4/0.3	0.0021	FP64, $< 10^{-4}$

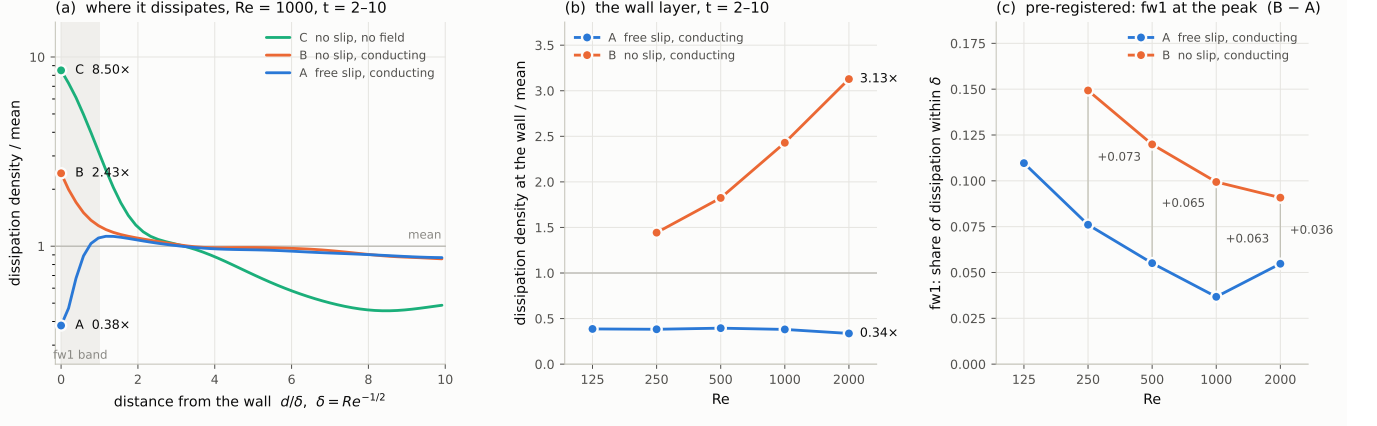


FIG. 1. Where the dissipation sits. (a) Dissipation density relative to the box mean against distance from the walls, d/δ with $\delta = Re^{-1/2}$, at $Re = 1000$, integrated over $t = 2-10$; the shaded layer is that of f_1 . (b) The density at the walls against Re , over the same window. (c) The pre-registered observable: f_1 at each run's turbulent dissipation peak, with B–A at each rung. Each point is from the finer grid of its rung (Table II).

times the two rungs' bands added. By the rule fixed before the $Re = 2000$ runs, the outcome is “cascade”.

E. The peak switches

Figure 5 shows why the last step is not what it seems. The dissipation of A has an early maximum near $t = 2.2$ and a later one at $t = 3.6-4.4$. At $Re = 500$ they are 5% apart; from $Re = 1000$ the later one is the larger, and A's peak time jumps from 2.23 to 4.06. B's peak moves the same way between $Re = 1000$ and 2000, from 2.34 to 4.53: at $Re = 2000$ its early rise is only a shoulder [Fig. 5(b)].

And f_1 does not vary slowly through either maximum. A's oscillates with a period near 1.3, between 0.022 and 0.060 at $Re = 2000$. B's falls from 0.32 at $t = 1.5$ to 0.06 at $t = 3.7$ as the start-up layers decay, then rises to about 0.11 [Fig. 5(c, d)]. The pre-registered reading therefore compares the two boxes at different times: at $Re = 1000$, B at $t = 2.34$ and A at $t = 4.06$. Between $Re = 1000$ and 2000 it moved from a time at which B–A, compared at equal times, is near 0.06 at both Reynolds numbers (0.059 and 0.064 at $t = 2.3$) to one at which it is near 0.03 at both (0.035 and 0.028 at $t = 4.5$). At equal times, B–A does not fall with Re up to $t \approx 4$: at

$t = 3.5$ it is 0.034–0.036 at every rung, and at $t = 2$ it rises, from 0.062 to 0.097. It falls only from $t \approx 4.5$ on: at $t = 8$, from 0.107 to 0.046.

None of the checks of Sec. III could see this. The gate and the Mach and precision controls each compare runs at one Re , and those switch together: A's peak is at $t = 4.05$ and 4.06 on the two grids at $Re = 1000$, and at 4.04 with u_0 halved.

F. The trend depends on when it is read

The time-integrated share (9) has no maximum to pick, but it needs a window. Table IV and Fig. 2(a) give B–A over five. Over $t = 2-6$, around the turbulent peaks ($t_p = 1.90-4.53$), B–A is flat: 0.049, 0.048, 0.051 and 0.048, within 6% at every rung. Windows reaching further into the decay give a falling trend, -18% over $t = 2-10$ and -43% over $t = 5-10$ from $Re = 250$ to 2000.

A fixed window compares different phases of the decay, since the evolution slows as Re grows: A's peak moves from $t = 2.0$ to 4.45. Aligning each run's window on its own energy, from the time when the total energy E_T has fallen to a fraction f_a of its initial value to the time when it has fallen to f_b , does not settle the question either. From $E_T/E_T(0) = 0.9$ to 0.5, B–A falls at every step,

TABLE III. Results on the finer grid of each rung. At the peak: t_p , the time of the turbulent dissipation peak, and B–A in f_1 there, the pre-registered observable. Integrated: the share F_1 of the dissipation over $t = 2-10$ within δ of the walls, and the viscous and Ohmic parts of B–A. The grid bands of the integrated B–A are at most 0.0009.

Re	at the peak (pre-registered)			integrated, $t = 2-10$				
	t_p^A	t_p^B	B–A	F^A	F^B	B–A	viscous	Ohmic
250	2.01	1.90	+0.073	0.069	0.128	+0.059	+0.072	−0.014
500	2.23	2.14	+0.065	0.056	0.111	+0.055	+0.069	−0.014
1000	4.06	2.34	+0.063	0.046	0.099	+0.054	+0.065	−0.012
2000	4.45	4.53	+0.036	0.038	0.086	+0.048	+0.059	−0.011

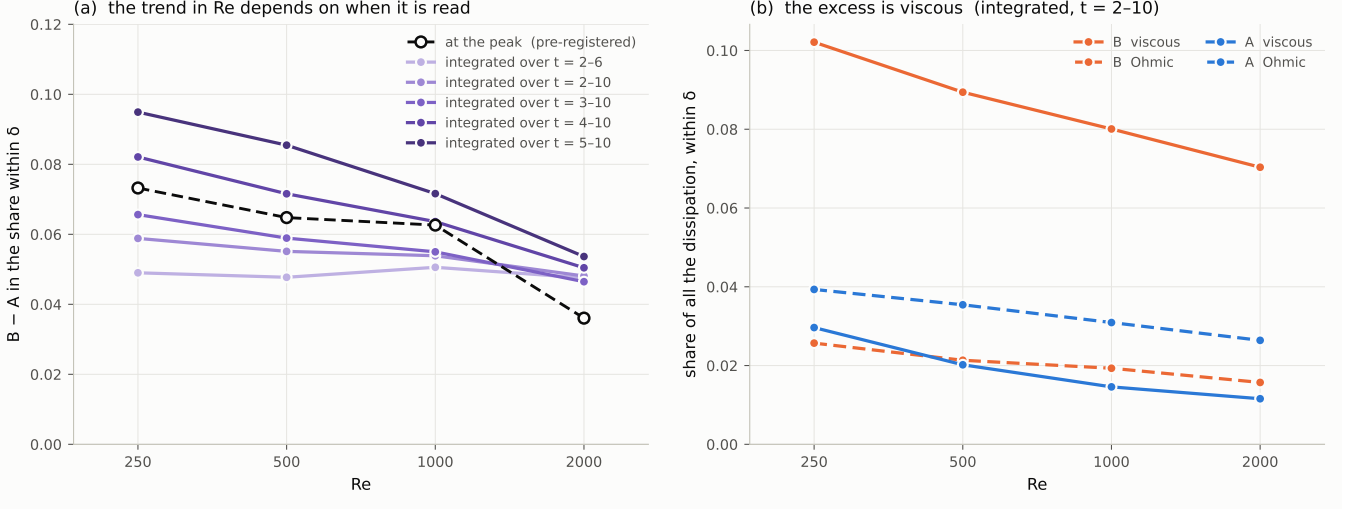


FIG. 2. (a) B–A in the share of the dissipation within δ of the walls against Re: at the turbulent peak, as pre-registered (dashed), and integrated over five time windows, after the fact (light to dark: later windows). (b) The integrated share F_1 over $t = 2-10$ for A and B, divided into its viscous and Ohmic parts. Grid bands are at most 0.001, smaller than the symbols.

from 0.166 to 0.048. From 0.6 to 0.4 it falls and then rises: 0.048, 0.029, 0.034 and 0.053. Applied to the default window $t = 2-10$, the registered rule also gives “cascade”, with B–A falling from Re = 1000 to 2000 by 0.0058 against bands of 0.0017. But the fall is 11% rather than 42%, and over $t = 2-6$ there is no fall at all.

G. Insulating walls

The pair A' , B' was meant to show whether the result needs a conducting wall. It cannot. A' passes the gate (worst 0.8%), and 88% of the dissipation within δ of its walls is Ohmic ($F_1 = 0.31$ and 0.27 at Re = 500 and 1000). B' has no turbulent peak by $t = 10$ at either Re. So $B' - A'$ (+0.32 and +0.42 over $t = 2-10$) compares a box that has not become turbulent with one that has, and under a different initial field from B–A. We draw no conclusion from it.

V. DISCUSSION

What the data settle. Every measure we tried agrees on four statements. First, a no-slip wall takes more of the dissipation next to it than a free-slip one. B–A is positive and far outside its band at every Re: at the peak, at every probe from $t = 1.5$ to 10 (where it is at least 0.019), and over every time and energy window. Second, the excess is viscous, and the near-wall Ohmic dissipation is lower, not higher, than at a free-slip conducting wall, by 40–45% in absolute terms. Third, the no-slip wall layer intensifies with Re: B’s wall dissipation density rises monotonically in every window, while A’s stays near 0.4 of the mean. Fourth, at the one rung where the hydrodynamic box is turbulent, the field moves dissipation away from the walls: the box without a field dissipates 33% of its energy within δ of the walls, the box with one 9.9%.

What they do not settle. The question that motivated the work was whether the wall share holds as Re grows, and it has a registered answer: “cascade”. That answer rests on the peak switching from B’s early maximum to its later one. A reading that does not depend on one instant depends instead on a window. B–A is flat over

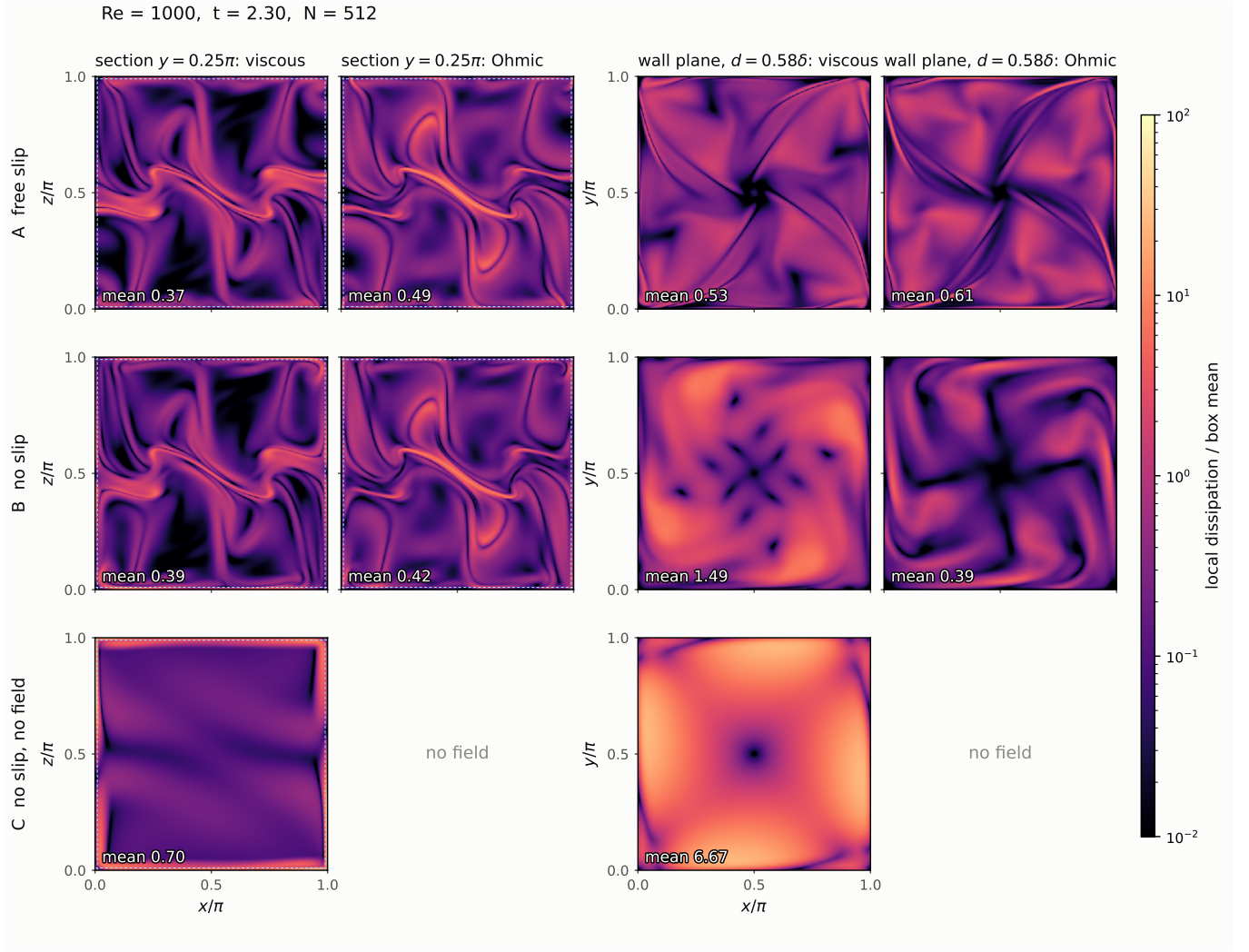


FIG. 3. Where the dissipation sits, at $Re = 1000$ and $t = 2.3$ (B's turbulent peak), on the finer grid ($N = 512$). Rows: the free-slip box A, the no-slip box B and the box without a field C. Columns: the viscous and Ohmic dissipation, $\nu|\omega|^2$ and $\eta|j|^2$, in the vertical section $y = \pi/4$, with dashed lines at δ from its four walls, and in the plane 0.58δ above the bottom wall. One logarithmic colour scale, relative to each box's mean dissipation at that time; each panel gives its plane's mean.

TABLE IV. B–A in the share within δ of the walls: at the turbulent peak, integrated over fixed time windows, and integrated over windows bounded by each run's own total energy. The last column is the change from $Re = 250$ to 1000. Grid bands are at most 0.0010 for the time windows and 0.0027 for the energy windows.

B–A in the share within δ	250	500	1000	2000	change (%)
at the peak (pre-registered)	0.073	0.065	0.063	0.036	–51
integrated, $t = 2$ –6	0.049	0.048	0.051	0.048	–3
integrated, $t = 2$ –10	0.059	0.055	0.054	0.048	–18
integrated, $t = 3$ –10	0.066	0.059	0.055	0.046	–29
integrated, $t = 4$ –10	0.082	0.072	0.064	0.050	–39
integrated, $t = 5$ –10	0.095	0.085	0.072	0.054	–43
integrated, $E/E_0 = 0.9$ –0.5	0.166	0.113	0.071	0.048	–71
integrated, $E/E_0 = 0.6$ –0.4	0.048	0.029	0.033	0.053	+11

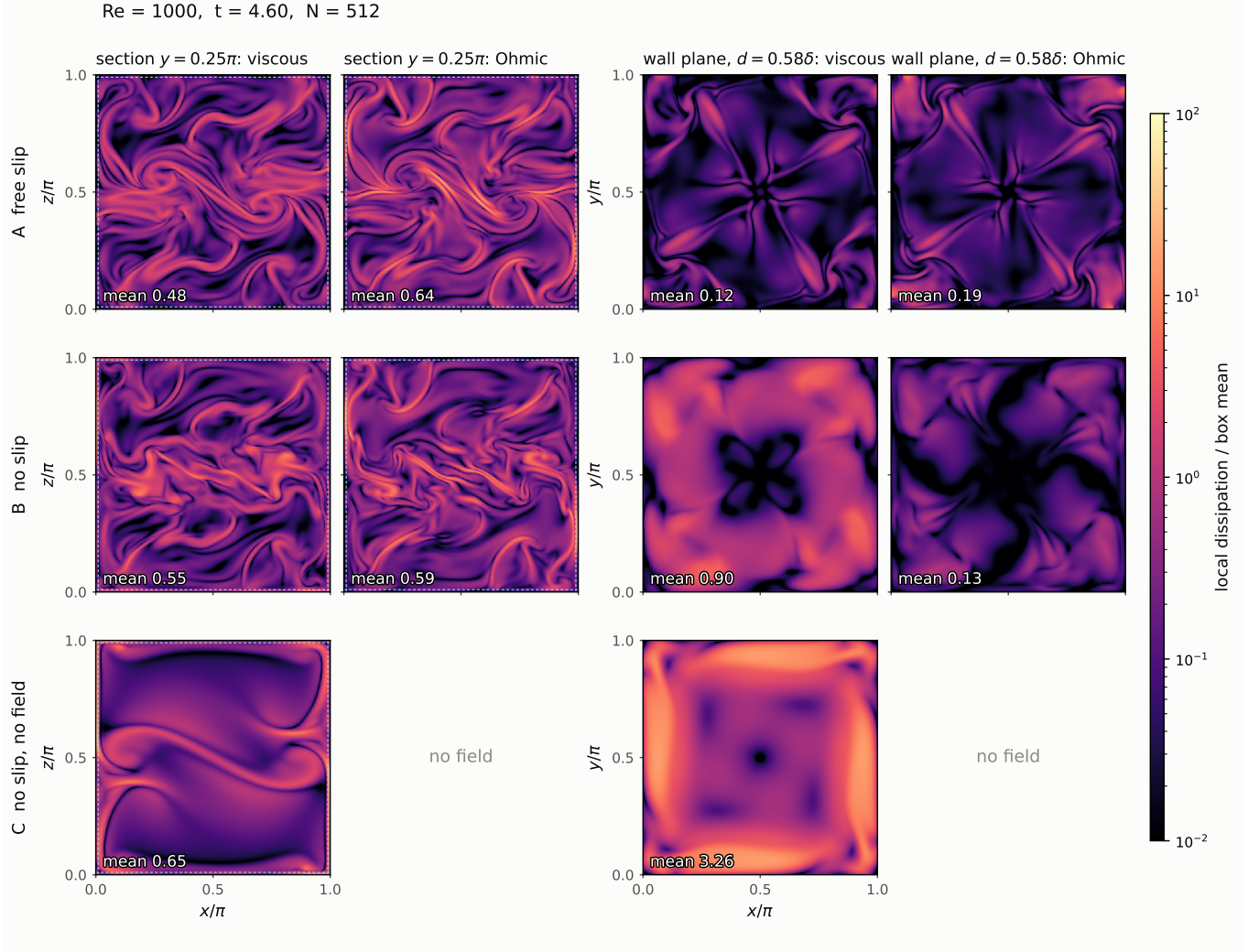


FIG. 4. As Fig. 3, at $t = 4.6$, by the later maxima of the dissipation.

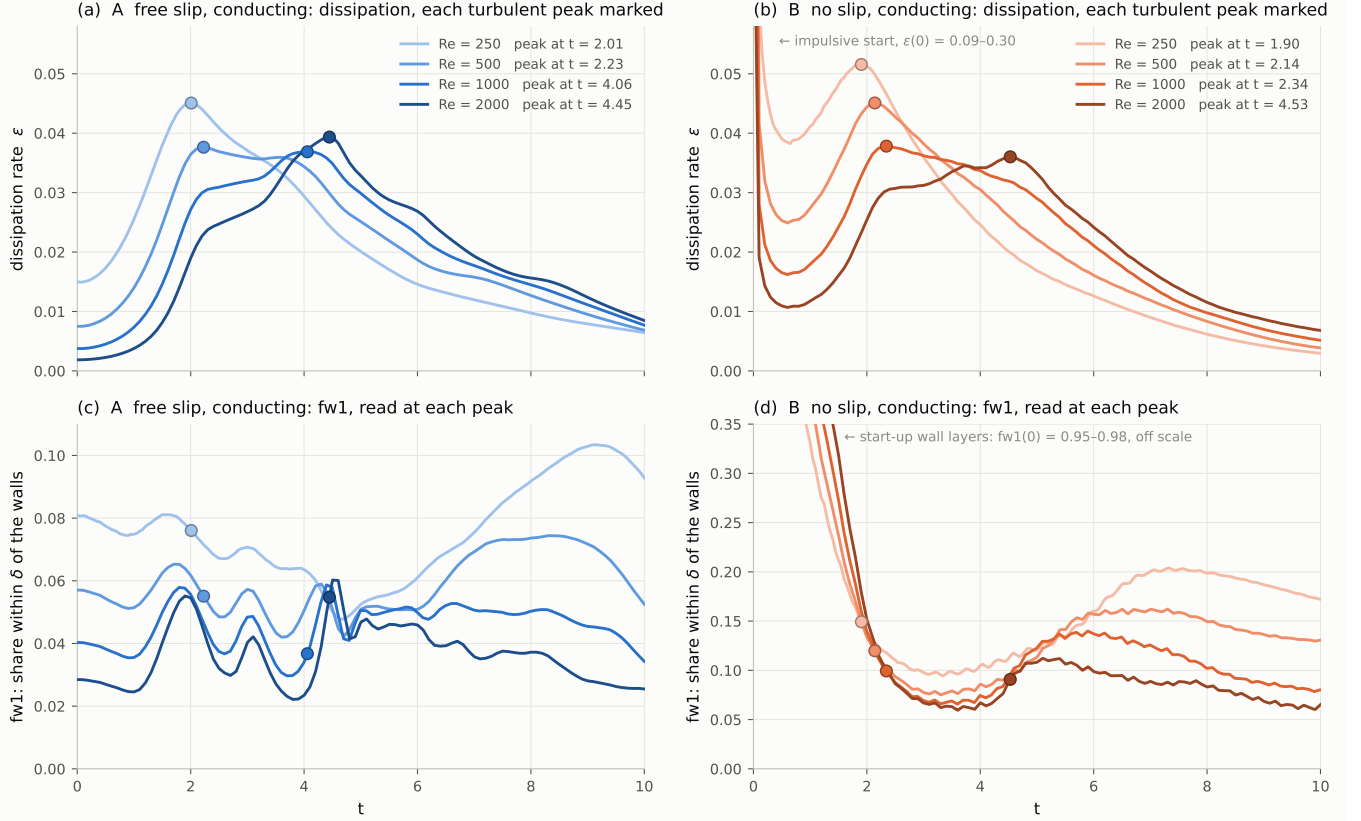


FIG. 5. The turbulent peak switches. The dissipation rate $\epsilon(t)$ (a, b) and the share within δ of the walls, $f_1(t)$ (c, d), for the free-slip box A (a, c) and the no-slip box B (b, d) at $Re = 250-2000$ (light to dark). Dots mark each run's turbulent peak and the f_1 read there. A's early maximum is the larger up to $Re = 500$ and its later one from $Re = 1000$; B's switches between $Re = 1000$ and 2000 . B starts off scale at $t = 0$ (impulsive start).

the window around the peaks and falls by 18–43% over windows that reach further into the decay, and aligning the windows on each run's energy moves the ambiguity rather than removing it. Over a factor of eight in Re , the trend of a decaying flow's wall share cannot be separated from the dependence of that share on the phase of the decay. We therefore claim neither outcome. The cascade verdict stands as the result of the test we registered, and we record that equally reasonable readings would not have produced it.

The layer we measure has the Prandtl thickness $\delta \propto Re^{-1/2}$, not Kato's $\propto Re^{-1.7}$. Over $t = 2-10$, B's share within δ falls as $Re^{-0.19}$ and A's as $Re^{-0.29}$, both more slowly than the volume of the layer, which falls as $Re^{-1/2}$. Whether a finite part of the dissipation remains at the walls as $Re \rightarrow \infty$ cannot be decided from these data.

A conjecture for the Ohmic deficit. At a free-slip conducting wall the tangential flow reaches the wall, and with $\mathbf{B} \cdot \mathbf{n} = 0$ and $\partial_n \mathbf{B}_t = 0$ it can stretch tangential field along it, building current sheets next to the wall. In A the near-wall dissipation is mostly Ohmic, and the snapshots of Sec. IV C show current sheets crossing the plane 0.58δ above the wall. A no-slip wall stops the tan-

gential flow, and with it the induction in the layer: in B that plane carries 0.68–0.79 of A's Ohmic dissipation and several times its viscous dissipation. The snapshots are consistent with the conjecture but do not test its mechanism, which needs the induction term itself in the layer, and they show one class of walls at two instants.

What the protocol bought. Had the decision rule not been fixed in advance, any window of Table IV could have been chosen after seeing the data. Over $t = 2-6$ it would have supported “walls hold”, over $t = 5-10$ “cascade”. Fixing the rule did not make the observable robust, and it was not. But it made the fragility something to report rather than something to choose around.¹³ In a decaying flow a quantity read “at the peak” compares runs at different times whenever the dissipation has more than one maximum of similar height, and TG flows, with their sequences of sheet formation and collision,^{2,3} are such flows. Before reading a trend from one instant of a transient, the time series at every parameter value should be inspected, and more than one definition of “when” should be tried.

Limitations. One initial condition, of high symmetry, in one realization; unit magnetic Prandtl number; and $Re \leq 2000$, set by the memory of one GPU, since

resolving δ with about 4.5 spacings makes N grow as $\text{Re}^{1/2}$. The method is weakly compressible. The Mach control bounds its effect at $\text{Re} \leq 1000$, but it was not repeated at $\text{Re} = 2000$, where the lattice Mach number is largest (0.133). The regularized edges and corners do not conserve mass exactly, though the drift stays below 4.4×10^{-5} . A forced, statistically stationary box would remove the ambiguity of “when” altogether, since the share would become a time average with a standard error. It is the natural next experiment, with its own decision rule fixed before it runs.

VI. CONCLUSIONS

We isolated what no-slip walls do to decaying MHD turbulence by solving the Taylor–Green MHD flow in its closed fundamental box with free-slip and no-slip walls and nothing else changed, and we fixed the observable and the decision rule before the runs. No-slip walls add dissipation in a layer $\text{Re}^{-1/2}$ thick at every Reynolds number from 250 to 2000. The excess is viscous and comes with less Ohmic dissipation at the wall, and the wall layer intensifies as Re grows. Snapshots of the local dissipation show the same contrast in space: current sheets cross the near-wall layer of the free-slip box, and a broad viscous layer, with weaker currents, lines the no-slip walls. The pre-registered reading, the excess at the dissipation peak, falls between $\text{Re} = 1000$ and 2000, which is the outcome we had called “cascade”. It does so because the peak switches from one maximum of the dissipation to another. Over this range of Re the trend of the wall share depends on when it is measured, and its sign does not.

ACKNOWLEDGMENTS

The computations used the Computational Shared Facility (CSF3) of The University of Manchester.

AUTHOR DECLARATIONS

Conflict of interest

The author has no conflicts to disclose.

DATA AVAILABILITY

The code (both implementations), the job scripts with the decision rules fixed before the runs, the time series and wall-distance profiles of every run, the plane means of the field snapshots, the analysis tools, and the script that produces every number in this paper are in the M3LB repository, <https://github.com/alexderosis/>

M3LB. The field dumps behind Figs. 3 and 4 (11 GB per run) are available from the author.

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